A Pitch-Synchronous Study of Formants

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Summary: Introduction. Formants are of fundamental importance in voice science. To date, formants have typically been studied using pitch-asynchronous methods, such as linear-prediction analysis. The results are often incomplete (without level), not objective (with frequencies depending on the preset order p), and require many pitch periods of stationary signals. A method that is accurate, complete, reproducible, and widely applicable is needed.

Method. This study presents a pitch-synchronous method for measuring formants. From the waveform of each pitch period, formants are obtained with high reproducibility, including all formant parameters such as central frequency, level, and bandwidth.

Results. The method was tested on 78 utterances of recorded sustained vowels with simultaneously acquired electroglottograph signals, segmented into 4730 individual pitch periods. For each waveform segment, Fourier analysis was applied to obtain an amplitude spectrum. Formants with three parameters were obtained from each amplitude spectrum. Using these formants, the voice waveforms were regenerated showing strong similarity to the original waveforms. The spectra can be averaged over many pitch periods to reduce noise and to estimate standard deviation.

Conclusions. Measuring formants from the waveform in each pitch period yields accurate, complete, and reproducible results. The method is applicable to live voices, including both speech and singing signals. The results can be used for voice research, speech and singing synthesis, and a quantitative study of phonetics.

Key Words: Voice—Speech—Singing—Vowels—Formants—Timbre.

INTRODUCTION

Formants are a fundamental concept in human voice research, encompassing both speech and singing. In speech research, vowels are typically characterized by a distinct set of formants.[1-3] Similarly, formants play a vital role in singing research.[4,5] Due to their significance in voice studies, a joint consensus paper involving 22 voice scientists was published in 2015 entitled *Toward a consensus on symbolic notation of harmonics, resonances, and formants in vocalization*.[6] The joint paper defines a formant as a property of the acoustic waveform of the radiated sound. Each formant is characterized by three parameters, central frequency F_n, level L_{Fn}, and bandwidth B_{Fn}.

The extraction of formant parameters from voice signals often employs linear-prediction coefficients (LPC).[7–9] However, LPC primarily provides the central frequency and the bandwidth for each formant, while the formant level is not available. Furthermore, the estimates of the central frequencies using LPC methods depend on the input parameter, the order p of the transfer function, see Figure 1. For instance, with $P = 4$, the LPC analysis detects only one formant, whereas $P = 8$ reveals three formants, while the one formant resulting from $P = 4$ is no longer present. This parameter dependency renders LPC neither reproducible nor objective.[2,9] Another formant estimation method is based on mel-frequency cepstral analysis (MFCC), widely used in speech recognition.[2,3] This method begins with a fast Fourier transform (FFT) on a windowed section of the voice signal, followed by a binning process on the frequency scale to flatten out the overtones of the fundamental frequency (see the top part of Figure 1). However, this method's accuracy is constrained by the sequence of binning frequencies and cannot estimate formant bandwidths. Both LPC and MFCC methods are based on a fixed processing window, typically 25 msec, multiplied by a window function, typically a Hamming or Gaussian window. Hence, a voice signal longer than 25 msec with a constant fundamental frequency is required.

THEORY AND METHOD

In this Section, we study the origin of formants from the point of view of voice production, then present a pitch-synchronous method for extracting formant parameters from voice signals in individual pitch periods.

The acoustic process of formant formation

In this subsection, we explore the acoustic process that leads to the creation of formants in the radiated waveform of a human voice. The starting point is the observed temporal correlation of voice signal and the electroglottograph signal, see the review articles of Baken and Herbst.[10,11]

A consistent experimental observation reveals a temporal correlation between the voice signal and the EGG signal. Each time the glottis closes, a strong impulse

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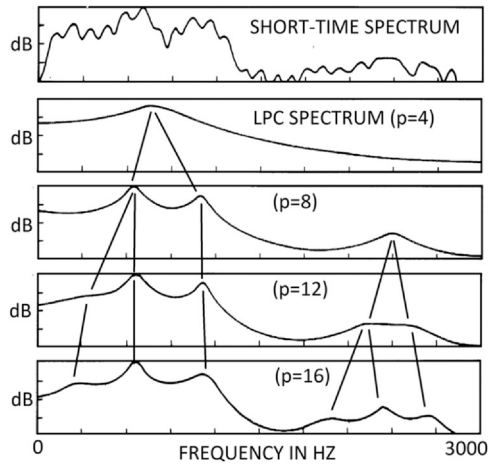

FIGURE 1. *Dependence of formant frequencies on* order *P of LPC.* To measure formants using LPC, the order of transfer function *P* must be entered first. With a different order *P*, the number of formants and their central frequencies change. Therefore, the LPC method of measuring formants is not objective. After Figure 8.18 on page 438 of Rabiner and Schafer.[2]

emerges in the voice waveform. The impulse subsequently oscillates and decays over time, as shown in Figure 2. The oscillation frequency is associated with the formation of *formants*, as detailed below. During the glottal closed phase, the oscillating acoustic wave decays at a constant rate. As the vocal folds decontact, air begins to flow past the glottis once again, and the next glottal closing event initiates a new decaying acoustic wave that overlaps with the remnants of previous acoustical waves.

The acoustic process of generating formants for a simple case of a uniform vocal tract is shown in Figure 3. During the glottal open phase, step (0), air flows from the trachea to the vocal tract. No sound is produced. An abrupt glottal closing, step (1), interrupts the air flow. However, the air flow in the vocal tract keeps moving due to inertia. A negative-pressure wavefront is generated, which propagates at the speed of sound toward the lips, then reflects from the lips and propagates toward the glottis. Because the glottis is closed, the wavefront is again reflected and propagates toward the lips. See Figure 3. A complete formant cycle takes four back-and-forth propagations over the length of the vocal tract at the speed of sound. The time of a complete cycle is

$$T = \frac{4L}{c}, \tag{1}$$

where c is the speed of sound, 350 m/sec, and L is the length of the vocal tract. For male speakers, $L \approx 17.5$ cm. Thus, $T = 2$ msec, or equivalently, a frequency of $F = 0.5$ kHz. If the vocal tract is a lossless tube of uniform cross section, a *square wave* of 0.5 kHz is produced. However, loss of energy is unavoidable. If the loss is small and frequency-independent, the Fourier series of square waves are

$$x(t) = Ce^{-2\pi Bt}\left[\sin 2\pi Ft + \frac{1}{3}\sin 6\pi Ft + \frac{1}{5}\sin 10\pi Ft + \frac{1}{7}\sin 14\pi Ft + \ldots\right], \tag{2}$$

which can be observed as a series of *formants* centered at frequencies 0.5, 1.5, 2.5, 3.5 kHz, and so on. The decay rate is characterized by a bandwidth B.

If the shape of the vocal tract is different from a uniform tube, the central frequencies and bandwidths of each formant change. In general, the waveform of an elementary sound wave triggered by a glottal closing is

$$\begin{aligned} x(t) &= \sum_{n=0}^{n<N} C_n e^{-2\pi B_n t}\sin 2\pi F_n t, \quad t \geq 0; \\ x(t) &= 0, \qquad\qquad\qquad\qquad\quad t < 0. \end{aligned} \tag{3}$$

The constant C_n is the amplitude, related to the level of the formant; B_n is the bandwidth, and F_n is the central frequency of the formant. Note that before a glottal closing, $t = 0$, there is no acoustic signal when the glottis is open. However, given that the glottal area during the open phase

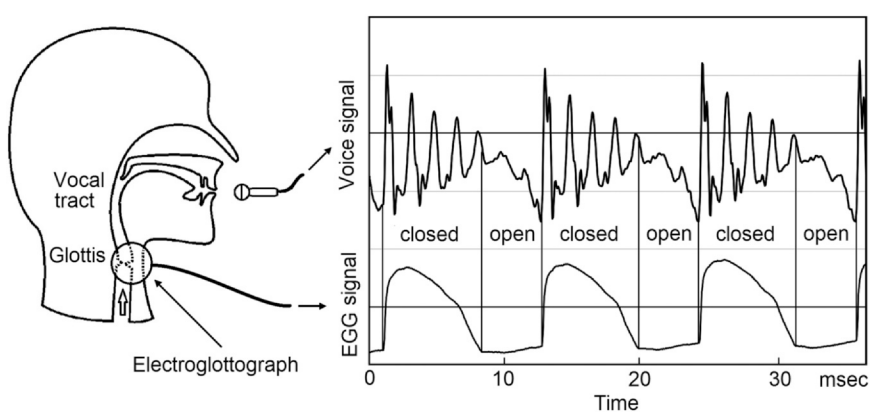

FIGURE 2. *Temporal correlation of voice and EGG signals.* At each glottal closing moment, a decaying elementary wave starts. A glottal closing blocks the air flow from the trachea to the vocal tract. As the flowing air in the vocal tract continues to move away from the glottis, a negative-pressure wavefront is generated, creating a sound wave. The sound wave propagates back and forth in the vocal tract. The complexity of that structure generates a cascade of rates of pressure change that results in the formation of a number of formants. See Figure 3. The time delay from the glottis to the microphone is corrected. After Miller.[12]

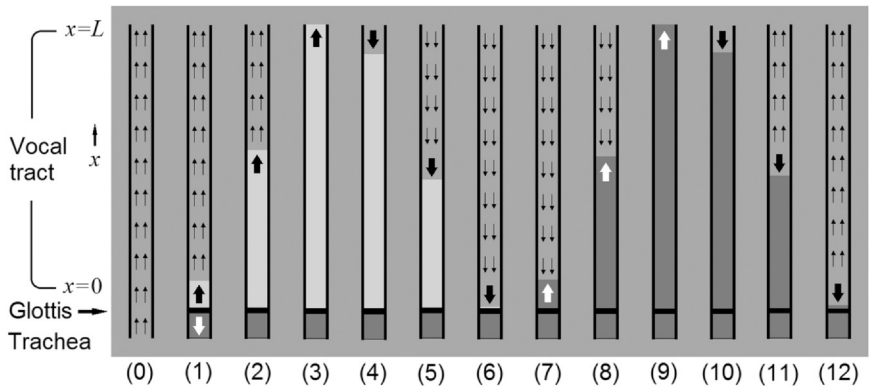

FIGURE 3. *Acoustical process of forming formants.* (0), while the glottis is open, air flows from the lungs to the vocal tract. No sound is produced. (1), a glottal closing blocks the air flow from the trachea to the vocal track. It creates a negative-pressure wavefront moving toward the lips at the speed of sound, (1) through (3). (4), the wavefront reflects at the lips. Because the pressure of open air is higher than the pressure in the vocal tract, air moves back, and the wavefront propagates toward the glottis, (4) through (6). (7), the wavefront reflects at the glottis to generate a wavefront of compressed air. (7) through (9), that wavefront propagates toward the lips. (10), the wavefront reflects at the lips. A wavefront of compressed air propagates toward the glottis, (10) through (12), to recover the initial state similar to (1).[13–15]

is much smaller than the mouth opening and the surface area of the vocal tract, the effect of the glottal area on the decaying wave is minor. Therefore, the decaying wave continues freely until it fully dissipates. Each subsequent glottal closure starts a new decaying wave superimposed on the tails of the residual waves from previous glottal cycles.

The concept of correlating formants with sound waveforms is not new. It was described by Peter Ladefoged in *Elements of Acoustic Phonetics*,[1] see Figure 4. For the vowel [o], as shown in Figure 4(A), the waveform can be understood as characterized by an impulse response dominated by a single decaying sinusoidal wave centered at

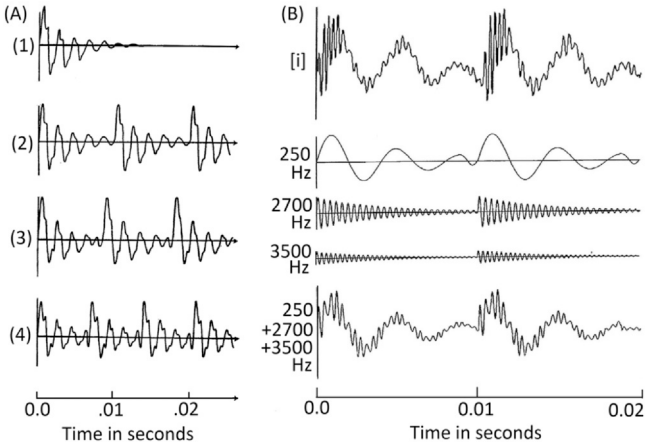

FIGURE 4. *Waveforms and formants of vowels.* **(A)** The sound waveform of vowel [o] can be understood as the superposition of a single decaying wave centered at 500 Hz at different repetition rates, which correspond to different pitch frequencies. **(B)** the waveform of vowel [i] can be understood by three decaying sinusoidal waves, centered at 250, 2700, and 3500 Hz, respectively. Adapted from Chapter 7, *The Production of Speech*, of Ladefoged's *Elements of Acoustics Phonetics* [1].

500 Hz. By superposing this decaying waveform at varying repetition rates corresponding to different pitch frequencies, different waveforms for the vowel [o] in (1) through (4) are explained. Figure 4(B) demonstrates that the waveform of the vowel [i] can be understood as a superposition of three decaying sinusoidal waves with frequencies of 250, 2700, and 3500 Hz, respectively. Repeating the three decaying waves with a repetition interval of 10 ms effectively replicates the sound waveform of vowel sound [i] at a pitch frequency 100 Hz.

In general, according to the transient theory of human voice production,[13–15] the acoustical wave of vowels is a superposition of elementary waves each triggered by a glottal closing and each carries full information of the timbre of the voice, thus termed a *timbron*. When the glottis is closed, the only energy available is the kinetic energy of moving air left in the vocal tract, which then resonates and ultimately radiates sound. The sound waveform in each pitch period is determined by the vocal tract geometry, and decays exponentially. Each individual decaying sinusoidal wave in Equation (3) can be considered a *formanton*, with each timbron consisting of multiple formantons, representing fundamental elements of the timbre of human voice.

Extracting formant parameters through discrete fourier transform (DFT)

The individual formantons overlap one another and are challenging to isolate. However, by applying a DFT, it becomes feasible to extract all three formant parameters from the sound waveform within a single pitch period, in most cases.

First, as expressed by Equation (3), the acoustical signal is zero before a glottal closing. If the properties of a few previous pitch periods are similar to the pitch period under investigation, then a DFT of the signals in the current pitch

period is equivalent to a DFT of the complete timbron over an infinitely long time. This is easily understandable from Equation (3) and Figure 4, as the waveform within the current pitch period contains the residual tails of the previous pitch periods, with the DFT computation incorporating this continuity.

Second, because the signal of a timbron is zero for $t < 0$, the phase spectrum of the waveform is uniquely determined by the amplitude spectrum because of the *dispersion relation*, or the Kramers-Kronig relation (see Section 5.1.1 of Chen, 2016).[15] Therefore, the amplitude spectrum alone provides sufficient information for analysis.

Third, the amplitude spectrum remains independent of the starting point in a pitch period. A mathematical proof can be found in Section 5.1.3 of Chen, 2016.[15] Therefore, the exact location of the segmentation point is inconsequential, provided it remains consistent across each pitch period.

Finally, an efficient method for performing DFT is the FFT. Historically, FFT algorithms required the input data array length to be a power of 2 (eg, 256, 512, or 1024). Because the number of the points in each pitch period is in general not a power of 2, the old-fashioned FFT cannot be applied without zero padding. However, using accessible modern-day programming languages, such as the FFT modules in NumPy, a standard package for Python, the length of the array can be any integer.

Because the timbron is zero before it is triggered by a glottal closing at $t = 0$, to make a Fourier analysis, the most convenient way is sine transform, defined as

$$A(f) = 2\int_0^\infty x(t)\sin 2\pi ft\, dt. \tag{4}$$

The reverse sine transform is

$$x(t) = 2\int_0^\infty A(f)\sin 2\pi ft\, df. \tag{5}$$

where $A(f)$ is the amplitude spectrum of $x(t)$.

The sine Fourier transform of each decaying sinusoidal wave in Equation (5) is

$$A_n(f) = 2\int_0^\infty C_n e^{-2\pi B_n t}\sin 2\pi ft\, \sin 2\pi F_n t\, dt$$
$$= \frac{C_n}{2\pi}\left[\frac{B_n}{B_n^2 + (f - F_n)^2} - \frac{B_n}{B_n^2 + (f + F_n)^2}\right]. \tag{6}$$

The overall spectrum is the sum of the spectra of all the formants,

$$A(f) = \sum_{n=0}^{n<N} A_n(f). \tag{7}$$

In the following, to simplify mathematics, an *independent formant approximation* is introduced. For typical (non-pathological) human voices, where the frequency distance between two adjacent formants is greater than the bandwidth, that model should work well. Following that model, within the frequency range around a formant frequency, the second term in Equation (6) is much smaller than the first term, and the terms of other formants in Equation (5) can be neglected. The peak value of amplitude spectrum $A_n(f)$ at the n-th formant is

$$A_n(F_n) = \frac{C_n}{2\pi B_n}. \tag{8}$$

Within the independent formant approximation, the second derivative of $A_n(f)$ evaluated at $f = F_n$ is

$$A_n''(F_n) = -\frac{C_n}{\pi B_n^3}. \tag{9}$$

The bandwidth can be evaluated as

$$B_n = \left|\frac{2A_n(F_n)}{A_n''(F_n)}\right|^{\frac{1}{2}}. \tag{10}$$

The second derivative of the amplitude spectrum can be evaluated by taking a Taylor expansion. Because the first derivative at a peak is zero, we have

$$A_n(F_n + \delta) = A_n(F_n) + \frac{1}{2!}A_n''(F_n)\,\delta^2 + \ldots. \tag{11}$$

The value of δ can be taken as, for example, 50 Hz. From the spectral curve, one can obtain an approximate value of the second derivative through a quantity V_n,

$$V_n = \frac{1}{\delta^2}\left[A_n(F_n) - \frac{A_n(F_n + \delta) + A_n(F_n - \delta)}{2}\right]$$
$$\approx -\frac{A_n''(F_n)}{2}. \tag{12}$$

Under that approximation, the bandwidth is

$$B_n = \left|\frac{A_n(F_n)}{V_n}\right|^{\frac{1}{2}}. \tag{13}$$

Within the independent formant approximation, a timbron can be considered as a collection of decaying waves in Equations (5). The energy of a timbron is the sum of the energies of individual decaying waves. For convenience, we may call a decaying sinusoidal wave associated with a formant a *formanton*. The energy of a formanton, up to a universal constant, can be computed as

$$E_n = \int_0^\infty C_n^2 e^{-4\pi b_n t}\sin^2 2\pi F_n t\, dt$$
$$= \frac{C_n^2}{8\pi B_n}$$
$$= \frac{\pi}{2}A_n^2(F_n)\, B_n. \tag{14}$$

Using the standard notation, the level or energy of the voice can be expressed by decibels (dB). Up to a constant, it is $10\log_{10}E_n$ dB. It is an important parameter. If the level of a formant is more than 40 dB lower than the level of the main formant, the energy of that formant is less than 1% of the total energy, which may or may not be audible, depending on the relative sensitivity of the human ear. In any case, we will focus on those formants that contribute more significantly to the total energy.

Waveform regeneration through superposition of formantons

The regeneration of waveforms based on the time-domain definition of formants is straightforward. According to Equation (3), a glottal closure at $t = 0$ triggers a timbron as a sum of formantons, each is a decaying sinusoidal acoustic wave,

$$x(t) = \sum_{n=0}^{n<N} C_n e^{-2\pi B_n t} \sin 2\pi F_n t. \tag{15}$$

To regenerate the voice waveform with a constant pitch, we note that the waveform in each pitch period is the sum of the timbron of the current pitch period and the tails of timbrons of the previous pitch periods

$$x(t) = \sum_{k=0}^{k<\infty} \sum_{n=0}^{n<N} C_n e^{-2\pi B_n (t+kT_0)} \sin 2\pi F_n (t + kT_0), \tag{16}$$

where T_0 is the pitch period. Because of the exponential factor, the sum over k converges quickly. Only a few terms are required.

SOURCE OF VOICE DATA

The pitch-synchronous formant measurement method was tested on a set of recorded voice samples with simultaneous electroglottograph (EGG) signals. The EGG signals were acquired using a D200 device from Laryngograph Ltd. Acoustic signals were captured with a DPA 4007 microphone (10 Hz-40 kHz, ± 2 dB) at a constant 25-cm distance from the singer's mouth opening, connected to a Metric Halo recording interface at 48-kHz 16-bit resolution. The whole recording chain was calibrated using a B-K acoustic calibrator for each specific microphone preamp gain setting. EGG signals were captured using two gold-plated electrodes placed over the singer's neck. Acoustic and EGG signals were synchronized and processed by the software package Speech Studio (Laryngograph Ltd., Wallington, UK). All collected samples were segmented to include sustained vowel portions only. Onsets and offsets were edited out using the Reaper software. The recordings were then compensated for gain to make up for differences in microphone preamp gains between different samples following the collected calibration data.

Voice samples from one professional singer were taken from a larger collection of voice samples to represent singing across the participant's pitch range in various dynamic and voice quality conditions. The male participant was 31 years of age at the time of study, had more than 25 years of singing experience, 12 of which at a professional level. The participant was an Authorised complete vocal technique (CVT) Teacher with 12 years of teaching and singing experience with the specific conditions in the protocol. CVT is a pedagogical and hierarchical framework for teaching professional voice.[16–22] At the time of study, the singer reported no vocal illness nor injury, had a Singing Voice Handicap Index of 3 with no self-reported problems with the singing voice. The singer was a bass-baritone voice with a physical range of C2 to F#6 based on a vocal range profile.

The participant was asked to sustain vowels for 2-3 seconds with two repetitions for each vowel on notes C3 (130 Hz), G3 (196 Hz), C4 (262 Hz), E4 (330 Hz), G4 (392 Hz), Bb4 (466 Hz), C5 (523 Hz), E5 (659 Hz), and G5 (784 Hz) in eight different conditions representing variations of 4 so-called vocal modes (the last two pitches only in the Falsetto condition). These vocal modes correspond to different vocal qualities of varying loudness with various vowels produced for each of the qualities. The conditions represent the vocal modes presented in CVT called Neutral, Curbing, Overdrive, and Edge,[16–21,23,24] in both a fuller and a reduced density condition as well as a Falsetto condition for the Neutral vocal mode. For a more detailed definition of the vocal qualities as well as previously reported norm data for them, the authors refer to the publications.[16–21,23,24]

For the Neutral vocal mode (*N*), vowel conditions included AH as in far [a], EE as in see [i], OO as in you [u], and EH as in stay [ɛ]. For the Overdrive vocal mode (O), vowel conditions included EH as in stay [ɛ] and OH as in so [ɜ]. For the Edge condition (E), vowel conditions included A as in and [æ], EH as in stay [ɛ], I as in sit [ɪ], and OE as in herb [ʊ]. Finally, for the Curbing vocal mode (C), vowel conditions included UH as in hungry [ʌ], O as in woman [o], and OO as in you [u]. For a more detailed description of audio-perceptual characteristics of the vocal modes, the authors refer the reader to Sadolin.[22] Samples were included for analysis in the study only after audio-perceptual confirmation by the participant in question and secondary confirmation by an Authorized CVT Teacher assessing both task accuracy and production comfort. Table 1 list the samples, with the vocal mode, vowel, measured pitch frequency, and number of pitch periods found in each sample that were adequate for analysis.

TABLE 1.
List of Voice Samples

Vocal Mode	Vowel	f_0 (Hz)	M
Edge	[æ]	503	41
Edge	[ɛ]	509	70
Edge	[ɪ]	510	43
Edge	[ʊ]	485	43
Overdrive	[ɛ]	512	86
Overdrive	[ɜ]	497	68
Curbing	[o]	516	36
Curbing	[u]	510	62
Curbing	[ʌ]	519	66
Neutral	[a]	513	95
Neutral	[i]	523	73
Neutral	[ɛ]	487	67
Neutral	[u]	522	53
Edge	[æ]	453	67
Edge	[ɛ]	450	93
Edge	[ɪ]	458	70
Edge	[ʊ]	459	69
Overdrive	[ɛ]	452	62
Overdrive	[ɜ]	455	96
Curbing	[o]	472	68
Curbing	[u]	460	56
Curbing	[ʌ]	462	53
Neutral	[a]	456	70
Neutral	[i]	469	78
Neutral	[ɛ]	461	93
Neutral	[u]	466	106
Edge	[æ]	386	69
Edge	[ɛ]	390	94
Edge	[ɪ]	385	72
Edge	[ʊ]	388	74
Overdrive	[ɛ]	384	65
Overdrive	[ɜ]	384	74
Curbing	[o]	387	92
Curbing	[u]	388	66
Curbing	[ʌ]	385	76
Neutral	[a]	387	84
Neutral	[i]	385	88
Neutral	[ɛ]	385	80
Neutral	[u]	388	78
Edge	[æ]	321	58
Edge	[ɛ]	323	72
Edge	[ɪ]	324	86
Edge	[ʊ]	257	52
Overdrive	[ɛ]	259	72
Overdrive	[ɜ]	260	68
Curbing	[o]	259	81
Curbing	[u]	259	73
Curbing	[ʌ]	263	47
Neutral	[a]	259	59
Neutral	[i]	261	59
Neutral	[ɛ]	258	62
Neutral	[u]	258	55
Edge	[æ]	190	59
Edge	[ɛ]	193	48
Edge	[ɪ]	193	50
Edge	[ʊ]	192	35
Overdrive	[ɛ]	195	38
Overdrive	[ɜ]	188	43

TABLE 1 (*Continued*)

Vocal Mode	Vowel	f_0 (Hz)	M
Curbing	[o]	194	64
Curbing	[u]	194	71
Curbing	[ʌ]	192	47
Neutral	[a]	192	52
Neutral	[i]	194	43
Neutral	[ɛ]	193	56
Neutral	[u]	195	53
Edge	[æ]	128	42
Edge	[ɛ]	128	32
Edge	[ɪ]	128	38
Edge	[ʊ]	127	38
Overdrive	[ɛ]	129	44
Overdrive	[ɜ]	128	70
Curbing	[o]	129	35
Curbing	[u]	129	43
Curbing	[ʌ]	130	31
Neutral	[a]	127	25
Neutral	[i]	128	41
Neutral	[ɛ]	129	29
Neutral	[u]	130	41

The average pitch value f_0 *of each sample is shown in Hz. In each sample, the number of pitch periods adequate for analysis are shown in the last column (M). For a description of the vocal modes and vowels, see* Section III.

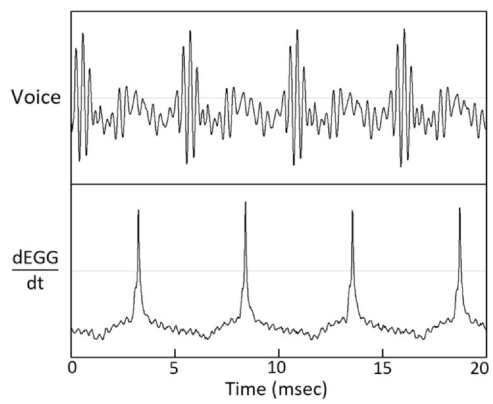

FIGURE 5. *Segmenting voice waveform using dEGG/dt data.* The dEGG/dt curve shows sharp and prominent peaks at the glottal closing instants. The time points of the peaks are utilized to segment the voice into pitch periods. Note: the audio waveform has not been time-corrected here.

DETAILS OF NUMERICAL PROCESSING

The original voice data were in the form of wave files of dual channels, a voice channel, and an EGG channel. The EGG channel takes a derivative to become dEGG/dt before further processing. The dEGG/dt curve shows sharp peaks at each glottal closing instant (Figure 5).[13-15] By identifying the positions of the sharp peaks in dEGG/dt signals, a series of segmentation points are generated. The voice signal is then segmented into pitch periods based on those segmentation points. At a 48 kHz sample rate, the accuracy of identifying the peaks in the dEGG/dt curve is

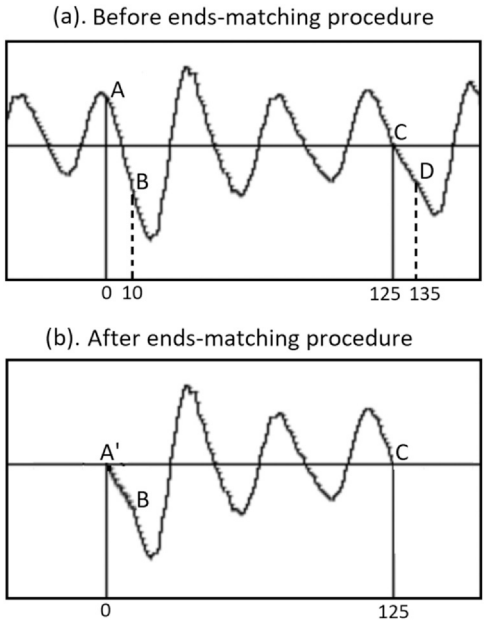

FIGURE 6. *The ends-matching procedure.* **(a)** The starting value A of a pitch period may be not equal to the ending value C. By taking more points after the end, for example, 10 points, CD, making a linear combination with the first 10 points AB and then replacing the first 10 points with this linear combination. **(b)** The beginning and the end of a pitch period become equal, making the waveform suitable for Fourier analysis.

about 0.02 msec. The accuracy of pitch-period segmentation is approximately 0.02 msec.

Because in general, right after segmentation, the value of the starting value A of the voice signal in a pitch period is not equal to the ending value C, an *ends-matching procedure* is executed, see Figure 6. For example, the original waveform in a pitch period is AC = wavin[125]. By taking 10 more points, CD = ext[10], using the Python code.

```
for n in range(10):

x[n] = 0.1*float(n)

wavin[n] = wavin[n]*x[n]+ext[n]*(1-x[n])
```

The beginning of the pitch period AB is replaced by the new section A′B. The ends are matched, and the curve is completely continuous, see Figure 6(b).

A key step is to take DFT on the waveform in each pitch period to generate spectra. In traditional FFT, the length of the data must be a power of 2, however, using the FFT modules in NumPy, the length of the array can be any integer. If the input sound wave is an array of wavin with n points, the code ampspec = np.abs(np.fft.rfft(wavin)) creates the amplitude spectrum of wavin containing (n//2 + 1) floating point numbers.

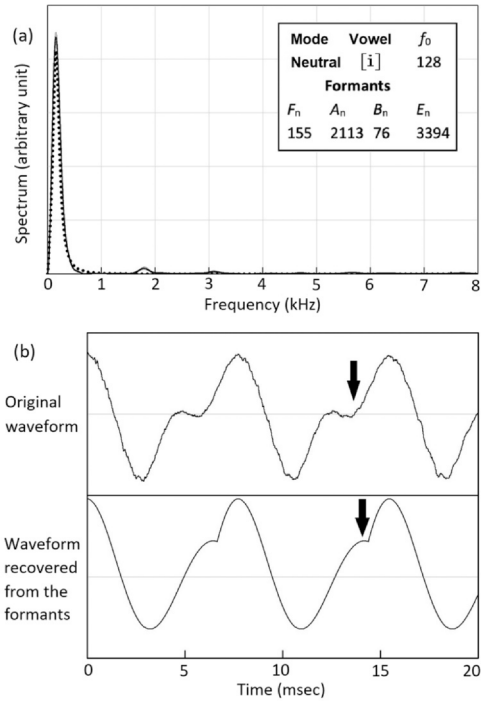

FIGURE 7. *Case of a low-frequency formant.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin error lines are not visible because the error is so small. Except for one formant at 155 Hz, the energy of all other peaks in the spectrum is less than 0.1% of the main formant. The glottal closing instant shows up as a kink in both the original waveform and the recovered waveform. Fn = frequency, An = amplitude, Bn = bandwidth, En = energy.

The next step is to interpolate the spectrum to a uniform frequency interval on a scale of the maximum frequency determined by the sampling theorem, for example, 2400 points on a scale of 24 kHz, with 10-Hz interval. This can also be done easily with Python.

With the amplitude spectrum represented as a series of 10 Hz intervals, the formants areidentified by searching the maxima of the spectral curve. By identifying three parameters of a peak (the frequency, the peak value, and the curvature of each formant), the three parameters for each formant (the central frequency, the level, and the bandwidth), can be obtained. Note that although the full spectrum is on a 24-kHz scale, all formant frequencies are lower than 8 kHz. To make the graphs visually pleasing, the frequency scale has been limited to representing 0 to 8 kHz.

An important feature of the current method is that data can be averaged and that standard deviation can be evaluated, since the amplitude spectrum can be obtained from each pitch period. The number of complete pitch periods in each utterance ranging from $M = 29$ to $M = 106$, Table 1. Consequently, a statistical study can be performed. At each

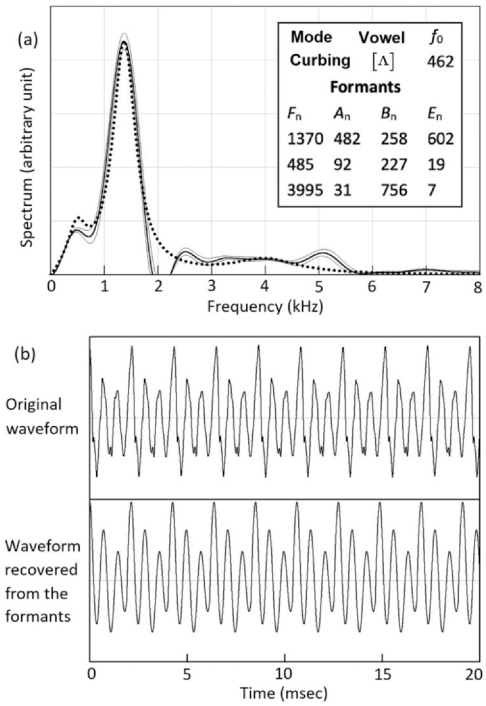

FIGURE 8. *Case of an integer ratio of frequencies.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin curves show the error. The dotted curve is the spectrum recovered from the formants. **(b)** In this example, the frequency of the dominant formant, 1370 Hz, is almost exactly three times the pitch frequency, 462 Hz. The waveform shows that with each pitch period, there are three formant cycles. It starts with a strong impulse, then gradually decays. There is no noticeable kink.

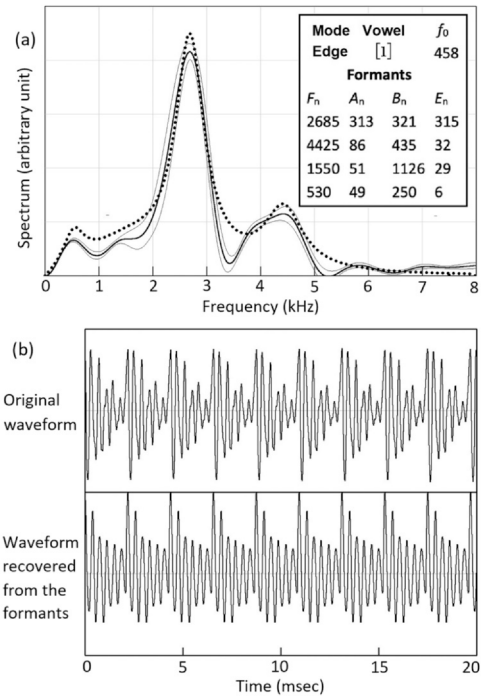

FIGURE 9. *Case of a strong high-frequency formant.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin curves show the error. The dotted curve is the spectrum recovered from the formants. **(b)** The original waveform and that recovered from the formants. The frequency of the main formant is 5.85 times the pitch frequency, showing almost 6 formant cycles in each pitch period.

frequency value f, each sample generates M values of amplitude Fourier coefficients $A_m(f)$, $0 < m < M$. Therefore, at each frequency value, there is an average amplitude spectrum

$$\overline{A(f)} = \frac{1}{M}\sum_{m=0}^{m<M} A_m(f) \tag{17}$$

and a standard deviation

$$\sigma(f) = \sqrt{\frac{1}{M}\sum_{m=0}^{m<M}[A_m(f) - \overline{A(f)}]^2}\,. \tag{18}$$

In Figure 7, an example of the amplitude spectrum is shown. The solid thick curve is the average amplitude spectrum $\overline{A(f)}$. The thin curves are $\overline{A(f)} + \sigma(f)$ and $\overline{A(f)} - \sigma(f)$, respectively. As shown, the deviation bar is narrow, indicating excellent consistency.

The formants are extracted using Equations (5) through (14). Although four peaks are observed in the spectrum, three of them have energy less than 1% of the total energy. Therefore, only one formant dominates the complex pattern and is focused on.

The correctness of choosing only one format in this case is validated by recovering the waveform. To do that, we need the pitch period T_0 as a parameter,

$$T_0 = \frac{1}{f_0}, \tag{19}$$

in this case it is 2.58 msec. Through Equations (15) and (16), the waveform is regenerated.

Figure 7 shows an example having a single low-frequency formant. The formant frequency, 155 Hz, is 1.21 times the pitch frequency, and dominates the voice energy. A new timbron starts at a time when the waveform of the preceding period is leveling. In both the original voice waveform and the recovered waveform, a kink is observed, as marked by an arrow.

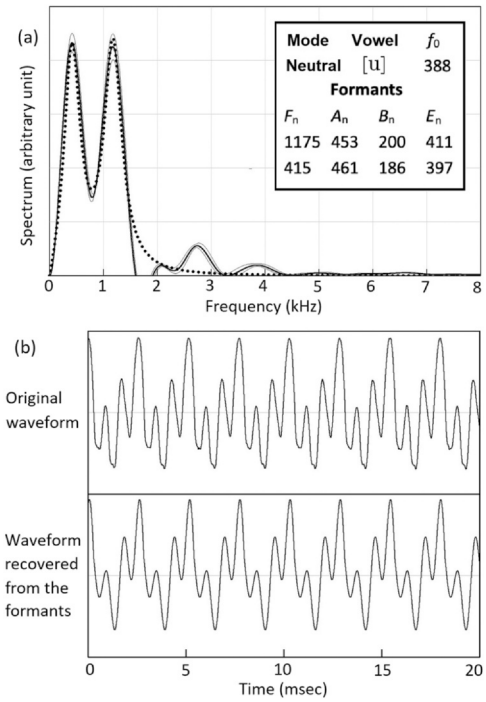

FIGURE 10. *Case of two formants of similar energy level.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin curves show the error. The dotted curve is the spectrum recovered from the formants. **(b)** The original waveform and that recovered from the formants. The frequency of the main formant is close to the pitch frequency, showing one structure in each pitch period.

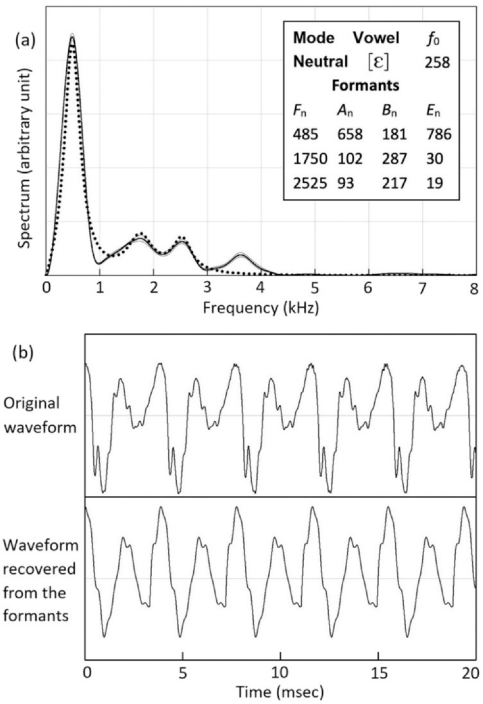

FIGURE 11. *Case of a strong low-frequency formant and weaker high-frequency formants.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin curves show the error bar. The dotted curve is the spectrum recovered from the formants. **(b)** The original waveform and that recovered from the formants. The frequency of the main formant is twice the pitch frequency, showing two formant cycles in each pitch period.

RESULTS AND DISCUSSIONS

The method was applied to all 78 utterances with 4730 individual pitch periods. For each voice waveform in an individual pitch period, Fourier analysis was applied to obtain an amplitude spectrum. Formants with three parameters were obtained from each amplitude spectrum. From the formants, the voice waveforms were regenerated, showing strong similarity to the original waveforms. The spectra are averaged over many pitch periods to reduce noise and to estimate standard deviation.

Figure 8 shows a case where the frequency of the dominant formant, 1370 Hz, is almost exactly three times the pitch frequency, 462 Hz. The waveform in each pitch period has three formant cycles. It starts with a strong impulse, then gradually decays. Because the starting point of a new timbron coincides with the rising edge of the preceding timbron, there is no noticeable kink. Except for minor oscillations at higher frequencies, the waveform recovered from the formants matches well with the original waveform.

In Figure 9, the frequency of the dominant formant is 5.85 times the pitch frequency. In both the original waveform and that recovered from the formants, each pitch period shows almost six formant cycles.

Figure 10 shows an interesting case of two formants with comparable energy, or level. The wave pattern created by the two formants is an alternation of a strong cycle and two weak cycles. The frequency of the lower formant almost matches the pitch frequency. The pattern generated by the two formants shows up in each pitch period. The waveform recovered from the formants is remarkably similar to the original waveform.

The case of a strong low-frequency formant and a bunch of weak high-frequency formants is shown in Figure 11. The central frequency of the main formant is almost twice as the pitch frequency. Therefore, the waveform shows two strong formant cycles in each pitch period, decorated with high-frequency features.

Figure 12 shows another case of a strong low-frequency formant and a group of weaker, high-frequency formants. In contrast with the case of Figure 11, the frequency of the main formant almost equals the pitch frequency, making one formant cycle for each pitch period. The

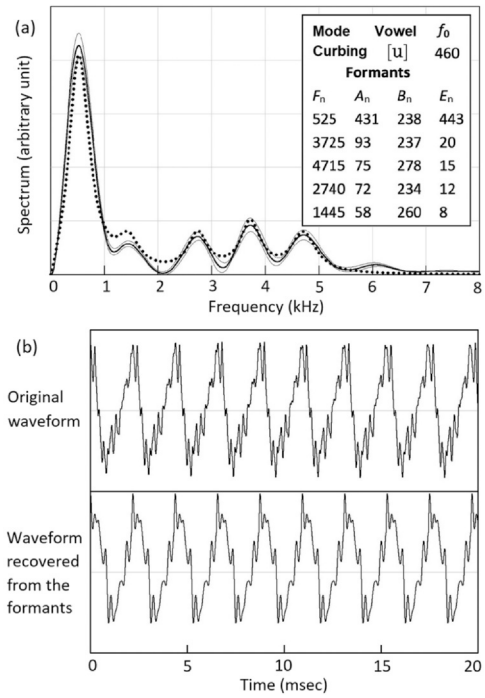

FIGURE 12. *Another case of a strong low-frequency formant and weaker high-frequency formants.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin curves show the error. The dotted curve is the spectrum recovered from the formants. **(b)** The original waveform and that recovered from the formants. The frequency of the main formant almost equals the pitch frequency, making a pattern of one formant cycle for each pitch period. The high-frequency formants are much stronger than those in Figure 11 and appear more prominently.

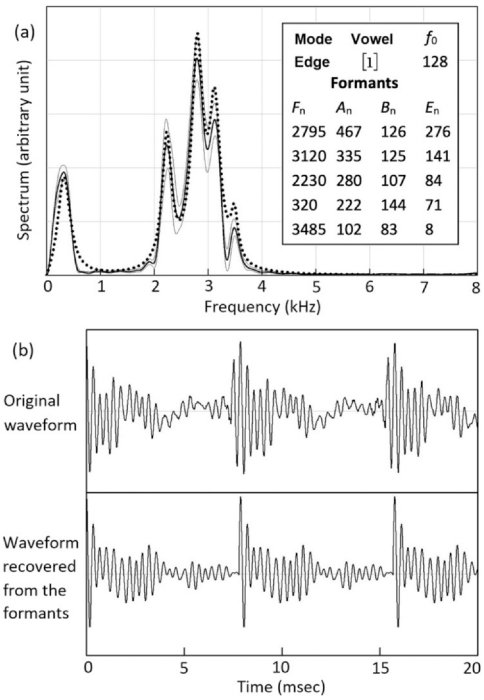

FIGURE 13. *Case of a weak low-frequency formant and group of strong high-frequency formants.* **(a)** The spectrum. The solid thick curve is the averaged amplitude spectrum $\overline{A(f)}$. The thin curves show the error. The dotted curve is the spectrum recovered from the formants. **(b)** The dominating feature is the very high-frequency oscillations related to the strong high-frequency formants. The weak low-frequency formant only generates a slight ripple.

high-frequency formants are stronger than those in Figure 11 and show up prominently.

The last example, Figure 13, shows a weak low-frequency formant and a group of strong high-frequency formants. This waveform is dominated by the high-frequency oscillations of those strong high-frequency formants, which ride atop the less-prominent, slower ripple of the low-frequency formant.

These examples serve to illustrate that a transient theory of formants allows for a detailed and nuanced analysis of different sounds with high accuracy between the formant-recovered waveform and the original sound waveform. This suggests that the methodology is accurate and reliable. Thus, the paper posits that a transient theory approach to formant analysis is objective, mathematically simple, and addresses many of the limitations of LPC and inverse filtering methods for estimating formant frequencies, including the pitch ranges typically limiting inverse filtering approaches. The method also improves accuracy and reproducibility as it is objective and does not require subjective input parameters.

CONCLUSIONS

A pitch-synchronous method to measure formants is presented. By applying it on a large set of voice samples, this study demonstrates that from the waveform in each pitch period, formants can be extracted with all parameters, including central frequency, level, and bandwidth, with high reproducibility. From the formants, the voice waveforms were regenerated showing strong similarity to the original waveforms, suggesting methodological accuracy. The method allows for averaging of spectra over many pitch periods to reduce noise and to estimate standard deviation. Future studies could utilize this approach to analyze in great detail the audio waveforms of different voicing conditions as well as pathological voice signals. Of particular interest to the singing community is the study of integer ratios between the fundamental frequency and formants (so called formant tuning) to generate either constructive or destructive interference for various pitch, vowel, loudness, and sound quality conditions. This is especially relevant to singing populations exploring efficient voicing at higher fundamental frequencies and intensitites. This warrants further study.

While this method was shown to work using sustained vowel recordings with nearly constant pitch and simultaneously acquired EGG signals, additional steps are required for live voice signals without EGG signals. In such cases, glottal closing moments would have to be extracted by using the method of multiplying the signal with an antisymmetric window function.[14,15]

The method explored in this study has broad applications in voice research, speech and singing synthesis, and quantitative phonetic analysis. This approach could also contribute to a quantitative definition of IPA vowels and potentially serve as the foundation for IPA-based voice synthesis.

Statement of Ethics

All procedures performed in this study were conducted in accordance with the World Medical Association Declaration of Helsinki. The professional singer gave written, informed consent for the data to be collected, analyzed, and disseminated in scientific research. The study was exempt from ethical approval in accordance with national guidelines in the Danish LBK nr 1338, §14, part 3 requiring only written, informed consent from the participant when data were anonymized at source.

Data Availability Statement

The data generated, extracted, and analyzed for this study are not publicly available to protect the privacy and anonymity of the professional singers who participated. Analytical extractions from the source data can be made available upon reasonable request to the corresponding author.

Declaration of Competing Interest

Author MA was employed in a Postdoc project administrated in part by Nottingham University Hospitals and in part by Complete Vocal Institute during data collection. The Postdoc project was supported by financing from the Danish Innovation Foundation (ref. no. 8054-00039B), which paid author MA a postdoc salary. The funder played no role in the design, data collection, data analysis, nor reporting of the study. Authors MA and IM are Authorized CVT Teachers. Authors AJ, IH, and JC have no conflicts of interest to declare.

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